

University of Asia Pacific

Department of Computer Science and Engineering

CSE 450: VLSI Lab

LAB REPORT

Experiment Number: **04**

Experiment Title: **Design and Simulation of an 8-Bit Register Using NAND Gates and Flip-Flops**

Submitted by:

Name: **MD Rawnak Shalahen Fahim**

Student ID: **22201063**

Section: **C1**

Submitted to:

Md. Rasheduzzaman
Lecturer
Department of Computer Science and Engineering

Date of Submission: **10/25/2026**

1. Title of the Experiment

Design and Simulation of an 8-Bit Register Using NAND Gates and Flip-Flops

2. Objectives

The objectives of this experiment are:

1. To design an **8-bit register** using structural VHDL.
2. To construct the register using **flip-flop modules**.
3. To implement the required flip-flop logic using **NAND gates**.
4. To understand hierarchical and structural design in VHDL.
5. To create and analyze the **RTL schematic** of the 8-bit register using Xilinx ISE.
6. To simulate the designed register and verify its functionality using a testbench.
7. To understand how individual flip-flops can be combined to construct a multi-bit storage element.

3. Software and Tools Used

The following tools were used for this experiment:

- **Xilinx ISE 14.7**
- **VHDL**
- **ISim Simulator**
- Computer system for design and simulation

4. Design Methodology

The 8-bit register was designed using a **hierarchical structural approach**.

Instead of directly describing an 8-bit register using behavioral VHDL, the design was divided into smaller modules. The basic hierarchy is:

NAND Gate → Flip-Flop → 4-Bit Register → 8-Bit Register

The complete 8-bit register consists of two 4-bit register sections:

- REG_LOW – stores the lower 4 bits
- REG_HIGH – stores the upper 4 bits

Each 4-bit register consists of four flip-flops, while the flip-flop is constructed using NAND gates.

Therefore, the overall structure can be represented as:

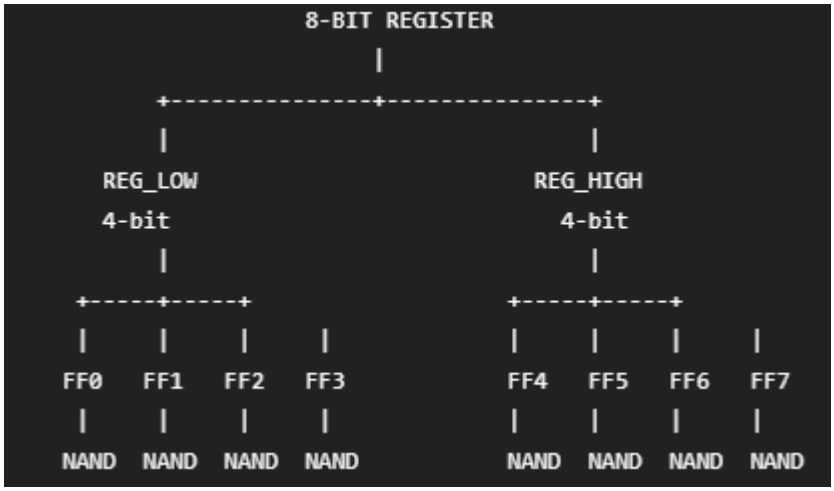


Figure 1: Overall hierarchical structure of the 8-bit register

Use the screenshot you provided showing:

- register
- register_8bit
- REG_LOW
- REG_HIGH
- flip-flops
- NAND gates

5. RTL-Schematic Diagrams

5.1 Top-Level RTL Schematic

The top-level RTL schematic represents the complete 8-bit register and shows how the lower and upper 4-bit registers are connected.

The 8-bit register is divided into:

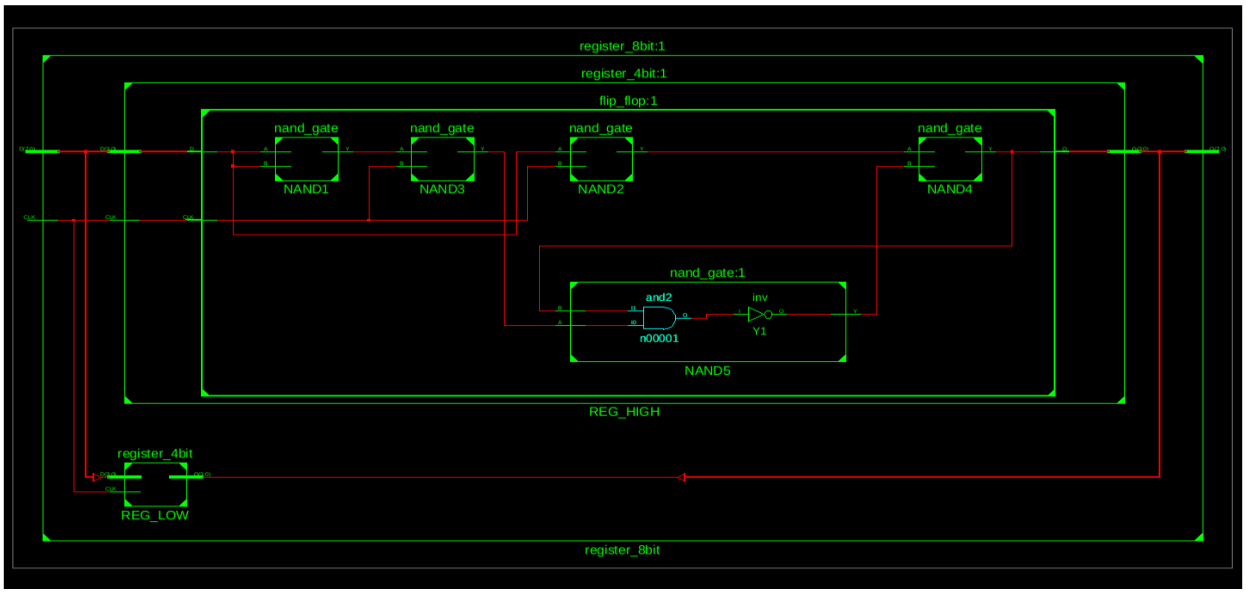


Figure 2: RTL schematic of the complete 8-bit register.

5.2 4-Bit Register RTL Schematic

Each 4-bit register is constructed from four flip-flops.

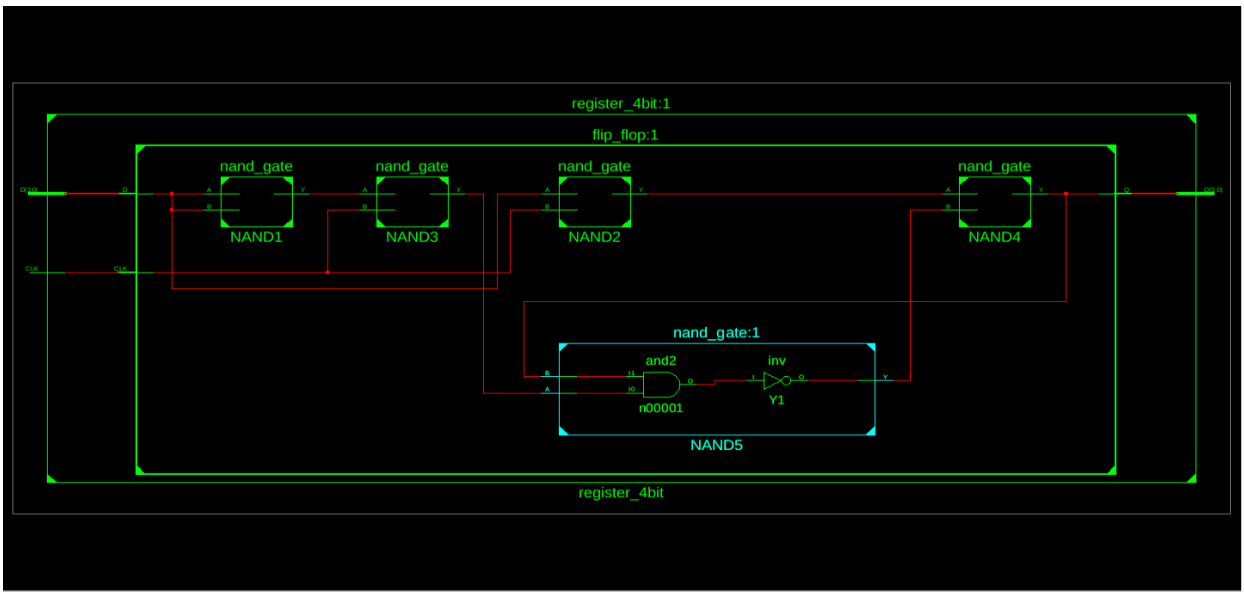


Figure 3: RTL schematic of the 4-bit register

The same structural design is used for both REG_LOW and REG_HIGH

6. Flip-Flop Design Using NAND Gates

The individual flip-flops were implemented using NAND gates.

The purpose of using NAND gates at this level is to demonstrate the construction of sequential logic from basic logic gates rather than using a pre-built register component.

The hierarchical structure of an individual flip-flop is:

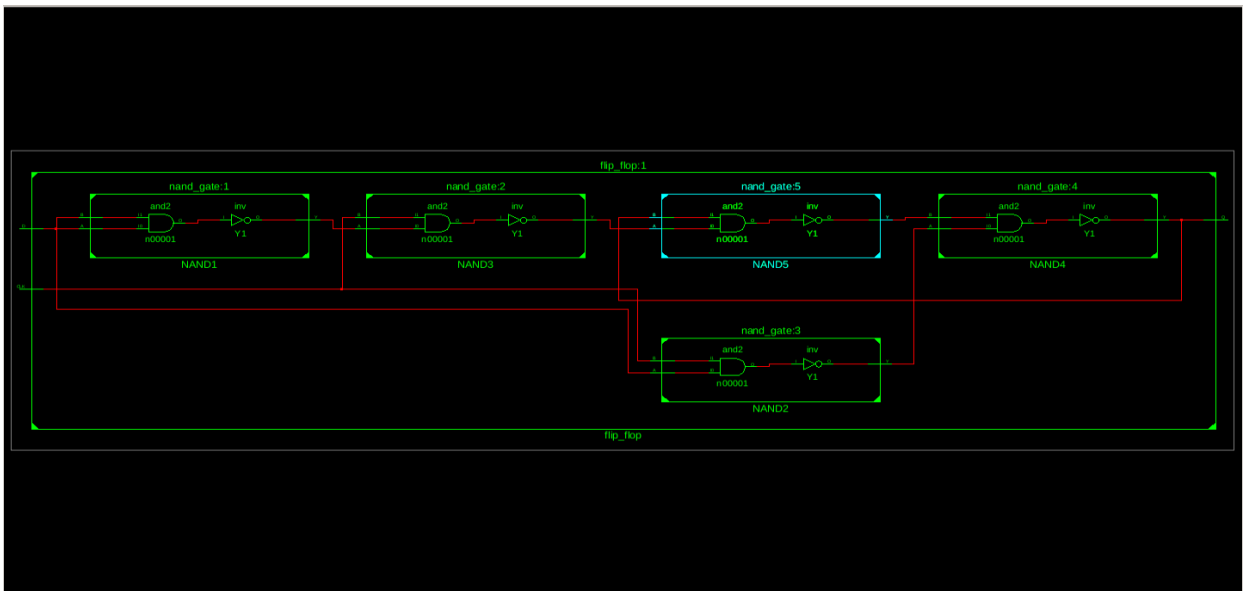


Figure 4: Internal NAND-gate implementation of a flip-flop

The NAND gates provide the basic logic required for storing the binary state.

In the Xilinx ISE hierarchy, the flip-flop modules can be expanded to observe their internal NAND-gate implementation.

7. NAND Gate Implementation

The NAND gate is used as the fundamental logic component in the flip-flop design.

The logical operation of a NAND gate is:

$$Y = \overline{A \cdot B}$$

Its truth table is:

A	B	Y = NAND(A,B)
0	0	1
0	1	1
1	0	1
1	1	0

Multiple NAND gates are interconnected to construct the flip-flop.

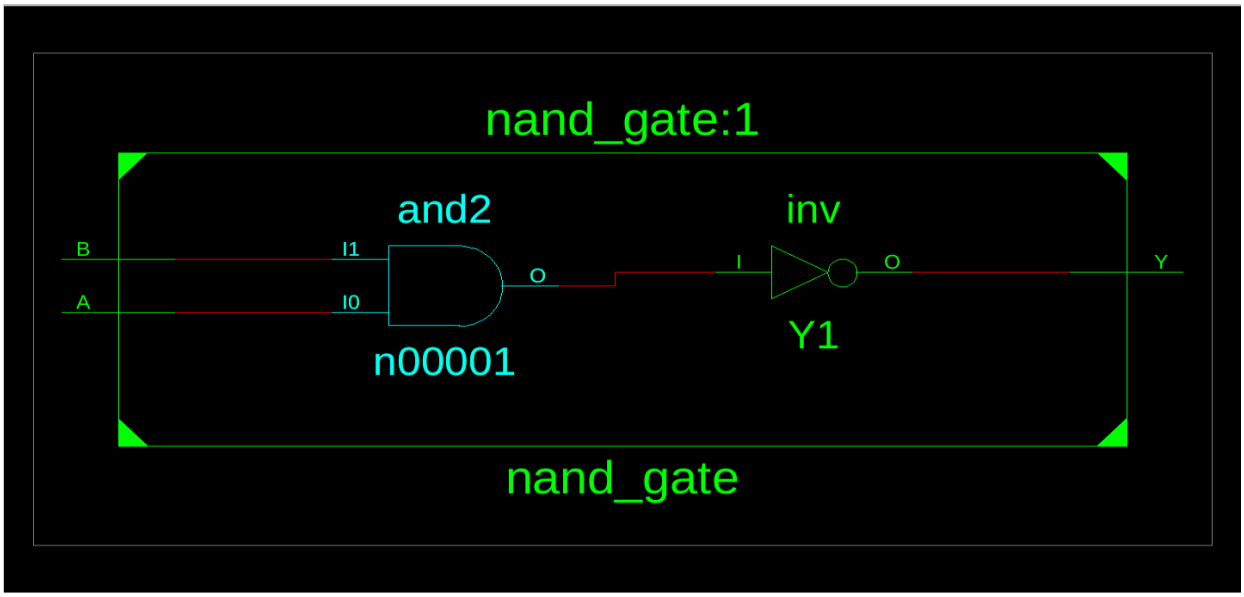


Figure 5: NAND gate used as the basic building block

8. OR Gate Using NAND Gates

The complete design was implemented using **structural VHDL**.

The design contains separate VHDL modules for:

1. NAND gate
2. Flip-flop
3. 4-bit register
4. 8-bit register
5. Testbench

The modules are instantiated hierarchically.

The basic design relationship is:



This modular approach makes the design easier to understand, simulate, test, and modify.

9. GitHub Repository

All VHDL source codes, testbenches, and related project files should be uploaded to a GitHub repository.

GitHub Repository Link:

https://github.com/rsf-rawnak/CSE450-VLSI-Labs/tree/main/register

10. Simulation

The designed 8-bit register was simulated in **Xilinx ISE ISim** using a VHDL testbench.
The testbench provides input signals to the register and observes the resulting output.
The main purpose of the simulation is to verify that:

- The register accepts 8-bit input data.
- The data is stored correctly.
- The outputs correspond to the stored data.
- The flip-flops operate correctly.
- The complete hierarchical design behaves as expected.

10.1 Simulation Waveform

The simulation waveform should demonstrate different input combinations being applied to the register.
For example:

Input 10101010
↓
Register : Stores the input
↓

Output 10101010

Another input can then be applied to verify that the register updates correctly according to the control/clock operation defined in your design.

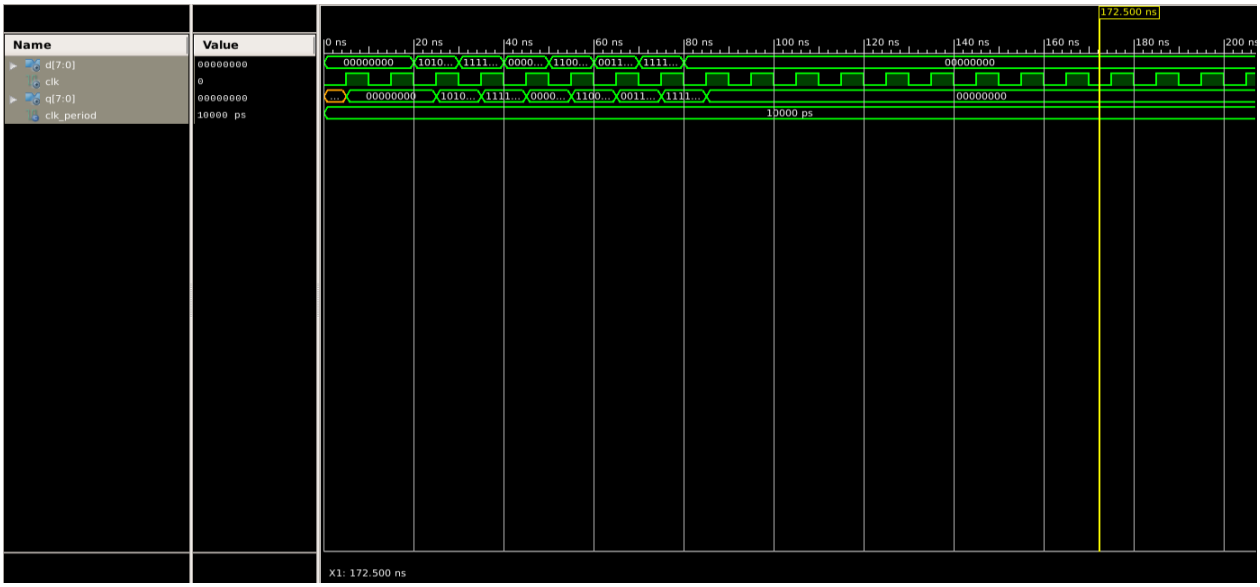


Figure 6: ISim simulation waveform of the 8-bit register

Open:

Xilinx ISE → Simulation → ISim

Run your testbench and capture the waveform showing:

- Clock
- Input bits
- Output bits
- Any reset/control signal you used

11. Simulation Analysis

The simulation waveform can be analyzed by comparing the input and output signals at the appropriate clock/control transitions.

When valid input data is applied and the required clock/control condition occurs, the flip-flops store the corresponding binary values.

For example, if the input is:

D[7:0] = 10110101

the register should store and produce the corresponding output:

Q[7:0] = 10110101

The simulation therefore verifies the data-storage behavior of the complete 8-bit register.

Important: In this paragraph, use the **actual binary values visible in your waveform** rather than the example **10110101** if your simulation uses different values.

12. Conclusion

In this experiment, an **8-bit register was designed and simulated using structural VHDL in Xilinx ISE**. The register was constructed hierarchically from smaller digital components.

The fundamental building block was the **NAND gate**, which was used to implement the flip-flop. Multiple flip-flops were then combined to form 4-bit registers, and two 4-bit registers were combined to create the complete 8-bit register.

The RTL schematic and Xilinx ISE hierarchy demonstrated the internal structure of the design, while the simulation waveform was used to verify the functional behavior of the register.

The experiment provided practical understanding of **hierarchical structural VHDL design, NAND-based sequential logic, flip-flop construction, register organization, RTL synthesis, and digital circuit simulation**.